MILITARY SPECIFICATION

DEMONSTRATION REQUIREMENTS FOR HELICOPTERS

This amendment forms a part of Military Specification MIL-D-23222A, dated 18 March 1971, and is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Under paragraph requirements: Delete “moving picture 3.1.6.1” and substitute “moving picture coverage 3.1.6.1.”

PAGE viii

Delete “demonstration instrumentation report 3.19.2.1" and substitute:

| | |
|---|---|
| Instrument Reports | 3.19.2.1 |
| Demonstration Instrumentation Report | 3.19.2.1.1 |
| Special Flight Test Instrumentation (SFTI) Inventory and Status Report | 3.19.2.1.2 |
| Special Flight Test Instrumentation (SFTI) Excess Equipment Report | 3.19.2.1.3 |
| Special Flight Test Instrumentation (SFTI) Requisition and Technical Data Report" | 3.19.2.1.4 |

PAGE 5

After paragraph 2.1.2, subparagraph NAVAIR Instruction: delete “13900.1A Special Flight-Test Instrumentation of the Naval Air Test Center; Accountability and Handling Procedures for."

PAGE 14

Paragraph 3.1.9.1, delete paragraph in its entirety and substitute:

“3.1.9.1 General. The kind and amount of Special Flight Test Instrumentation (SFTI), method of data acquisition and the information required by 3.19.2.1.1 necessary to comply with the demonstration require-ments of this specification, shall be proposed by the contractor for Naval Air Systems Command (NAVAIR) approval and subsequent inclusion as an Appendix to the addendum to this specification. The Naval Air Test

FSC 1520


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Center (NAVAIRTESTCEN) participation in review and verification of suitability of the proposed instrumentation will be required. By conferring with representatives of NAVAIRTESTCEN, the contractor shall make maximum utilization of Government-furnished SFTI as is available from the NAVAIR Special Flight Test Instrumentation Pool (SFTIP) at NAVAIRTESTCEN. All other required instrumentation shall be acquired by the contractor as contractor-acquired Government-owned SFTI. The contractor shall maintain repair, report, and administer Government-owned SFTI and when directed by COMNAVTESTCEN acting through the cognizant Contracting Officer, ship or release by incorporating "SF" as the last two characters of the item's model number. All old equipment, where practical, will be clearly marked with "SF".

PAGES 14 AND 15

Paragraph 3.1.9.2, delete paragraph in its entirety and substitute:

"3.1.9.2 Installation, calibration and maintenance. The contractor shall install and calibrate all aircraft instrumentation used in performing the demonstrations. All instruments and instrument systems shall be installed in accordance with the highest standards of mechanical and electrical installation practices. All transducers and strain-gage installations shall be properly located, shall be properly damped, shall have flat frequency response characteristics commensurate with the frequencies of excitation of the variable to be measured, and shall be properly mounted to assure valid measurements and freedom from extraneous excitations. The flat frequency response shall be not less than 60 Hz for all strain gages, pressure transducers, acceleration, velocity and displacement measuring instruments unless lower response characteristics, if proposed by the contractor, are concurred with by NAVAIR. Instrumentation recording systems including magnetic tape recorders and telemetry equipment, if installed and used on demonstration aircraft of NAVAIRTESTCEN, shall be compatible with the existing ground station equipment at NAVAIRTESTCEN. To the maximum extent possible, end-to-end calibrations of all parameters shall be made through the aircraft instrumentation system to at least the maximum range of excitation expected during the course of the demonstration. The calibration may be performed by a transducer substitution techniques for those transducers for which an accurate substitution model exists; however, this method is preferred only when an actual end-to-end calibration is not practical. Transferred laboratory calibrations shall be unacceptable, except in those cases in which the actual end-to-end calibration or transducer substitution methods are determined to be impossible or impractical. Calibration methods for all parameters shall be proposed to the Test Authority for approval. Calibration test data shall be obtained and recorded during both increasing and decreasing values of the pertinent parameter which the instrument is intended to measure, to insure repeatability and freedom from hysteresis.



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"All strain gage installations on simple and complex structures shall be installed to minimize interactions or 'cross-talk' during combined loadings; such interactions as do exist shall be properly accounted for during the calibration. Installation of strain gages which are impossible or impractical to calibrate shall be resorted to only if it can be shown, prior to such installation, that the computed loads from such installations are meaningful and useful, and provided further, that the methods of gage applications and load calculations from strain gage output, gage factor, and physical constants of the member are acceptable to the Test Authority. The instrumentation shall be operated and maintained by the contractor during the demonstration. If, during the course of the demonstration, it becomes apparent to the Test Authority that there is a change in the calibration of certain instruments, such instruments shall be recalibrated and their recalibration shall be witnessed and accepted in accordance with 3.1.9.4 and 3.1.9.5. Sufficient data (as a part of the report required by 3.19.2.1.1) shall be submitted by the contractor to the Test Authority for approval on the total airborne instrumentation system to insure (1) that no significant degradation of performance results from extraneous noise, vibration, shock or temperature, (2) that each channel has flat frequency response commensurate with the parameter to be measured, and (3) that errors resulting from time delays or phase shifts between measured parameters are commensurate with the accuracy requirements of the test program. A detailed description of all instrument and recording devices, methods of calibration, location of instruments, and calibration data for each demonstration aircraft shall be submitted as separate appendices to the report required by 3.19.2.1.1. Such appendices shall be submitted at least three weeks prior to the time the aircraft is scheduled to arrive at the test site."

PAGES 15 AND 16

Paragraph 3.1.9.3, delete paragraph in its entirety and substitute:

"3.1.9.3 Check-calibrations. Check-calibrations of all contractor installed instrumentation shall be performed or witnessed by a representative of the Test Authority before commencing a demonstration. Instrumentation system static and dynamic response checks shall be made through the recording systems telemetry transmitters and ground station equipment to assure that (i) freedom from extraneous noise exists, (2) that each channel has flat frequency response commensurate with the parameters to be measured, and (3) that the maximum errors resulting from time delays or phase shifts between measured parameters are within tolerance, and (4) that the system input/output is representative of the Parameter calibration. Where such check-calibrations show significant departures from previous calibrations, a complete recalibration shall be made in accordance with 3.1.9.4 herein."



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Paragraph 3.1.9.4: Delete "3.19.2.1" in the last sentence and substitute "3.19.2.1.1."

Paragraph 3.1.9.5, delete paragraph in its entirety and substitute:

"3.1.9.5 Acceptance and witnessing procedures. The contractor shall, as a minimum, evaluate SFTI under a simulated aircraft environment commensurate to that to be encountered during the aircraft test program. The evaluation shall be performed prior to the aircraft demonstration and witnessed by representatives of the Test Authority. Formal acceptance, by representatives of the Test Authority, of all instrumentation system installations for each demonstration aircraft shall be performed at the location at which the demonstration tests are to be performed, as specified in 3.1.1. In lieu thereof, and subject to the approval of COMNAVAIRTESTCEN, such acceptance may be performed at the contractor's facility or other designated location, at least two weeks prior to delivery of the aircraft at the test location, to preclude serious delay in the contractor's development program. Concurrent or prior to the foregoing acceptance of instrumentation installations, contractor calibrations of all instruments shall be witnessed by representatives of the Test Authority. The Test Authority may designate other Government representatives to witness instrumentation calibrations in accordance with the provisions of 3.1.2 Such witnesses shall adhere to the following minimum requirements:"

PAGE 17

paragraph 3.1.9.5, delete subparagraph (1) Installation in its entirety and substitute:

"(1) Installation. The installation of transducers, gages, signal conditioning and recording systems shall be reviewed and inspected to assure: (a) acceptable workmanship, proper location and mounting of all instruments and related systems; (b) that the signal conditioning equipment does not have any deleterious effects on measurement accuracy and data reduction; and (c) that instrumentation systems including magnetic tape recorders and telemetry equipment, if installed and intended to be used at NAVAIRTESTCEN, are compatible with NAVAIRTESTCEN ground station equipment."

PAGE 18

Paragraph 3.1.10, subparagraph (1): Delete "(3.19.2.1)" and substitute '"(3.19.2.1.1)."



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Paragraph 3.2.1, delete paragraph in its entirety and substitute:

"3.2.1 Instrumentation conference. To facilitate accomplishment of all demonstrations, conferences shall be held between representatives of the contractor and the cognizant Test Authorities. These conferences shall start as soon as practical but not later than 12 months prior to the scheduled release of the first aircraft for flight to (1) provide satisfactory lead times for procurement of instrumentation which, at the time of the conference, may not be available from the SFTIP as Government-furnished Special Flight Test Instrumentation (GFSFTI), and (2) to facilitate compliance with other provisions of this specification."

PAGE 95

Paragraph 3.19.1.1, format and general requirements: Delete entire subparagraph (8).

PAGE 96

Paragraph 3.19.1.3.1, add subparagraph (5):

"(5) Endorse SFTI Inventory and Status Reports, SFTI Excess Equipment Reports, and SFTI Requisition and Technical Data Reports to COMNAVAIRTESTCEN (TS241)."

PAGES 96 AND 97

Paragraph 3.19.13.2, delete paragraph in its entirety and substitute:

3.19.13.2 NAVAIRTESTCEN. Review Demonstration Instrumentation Reports, Demonstration Progress Reports, Avionics Systems Demonstration Data Reports, and Performance Data-Reduction Reports, and forward one copy of each to COMNAVAIR, attention of the organizational code having cognizance of the AIRTASK/Work Unit Assignment, with recommendation as to their suitability."

PAGE 97

Paragraph 3.19.2.1, delete title "Demonstration Instrumentation Report" and substitute: "3.19.2.1 Instrumentation Reports."

Add paragraph 3.19.2.1.1, 3.19.2.1.2, 3.19.2.1.3 and 3.19.2.1.4:

"3.19.2.1.1 Demonstration Instrumentation Report. This-report shall be submitted via COMNAVAIRTESTCEN not later than three months prior to the last date at which it will be practical to install in demonstration aircraft, items or components of test instrumentation which must be



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installed therein during manufacture. The report shall be sufficiently complete to indicate the need for the instrumentation in fulfilling the demonstration requirements, shall comply with the requirements of 3.1.9 and shall contain:

a. Complete list of the demonstration specification and/or its contract addendum for which each instrument will be used.

b. Complete list of variables to be measured with each demonstration aircraft and the expected overall accuracy of measurement of each variable; a discussion and listing of expected errors resulting from time delays or phase shifts between measured paragraphs.

c. *Complete list of Government-furnished and contractor-acquired SFTI.

d. Estimated dates for completing installation of instrumentation in each demonstration aircraft.

e. **Detailed description of all instrumentation and related systems and all final calibration data for each demonstration aircraft.

f. Sufficient data on the total instrumentation system to insure (1) that no significant degradation of performance results from extraneous noise, vibration, shock and temperature, (2) that each parameter has flat frequency response commensurate with the parameter to be measured, and (3) that expected efforts resulting from time delays or phase shifts between measured parameters are commensurate with the accuracy requirements of the test program.

* This list shall completely identify the instrumentation as to purpose, function intended, location and response characteristics required.

** May be submitted as a separate appendix to the report no later than three weeks prior to the time the aircraft is scheduled to arrive at the test site."

"3.19.2.1.2 Special Flight Test Instrumentation (SFTI) inventory and Status Report. This report shall provide the inventory listing and status of Government-furnished and contractor-acquired SFTI in custody of the contractor, and shall contain the serial number, nomenclature, model number, range, manufacturer, cost, date equipment was acquired, present location, aircraft or contract to which assigned, date equipment was assigned to this contract and current status of each item of SFTI."


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"3.19.2.1.3 Special Flight Test Instrumentation (SFTI) Excess Equipment Report. This report shall be submitted to COMNAVAIRTESTCEN at the time Government-owned SFTI becomes excess. This report shall contain the serial number, nomenclature, manufacturer's model number, range, manufacturer, cost, date equipment was acquired, present location, aircraft or contract to which assigned, data equipment was assigned to this contract and status of each item of excess SFTI."

"3.19.2.1.4 Special Flight Test Instrumentation (SFTI.) Requisition and Technical Data Report. This report shall include:

a. Requisition and Invoice/Shipping Document (DD Form 1149). This form shall contain the contract and program to which the SFTI is assigned, the nomenclature, model number, serial numbers range and date acquired for each item of SFTI, and the statement that, "The contractor shall maintain, repair, report and administer this Government-owned SFTI in conformance with sound industrial practice and in accordance with applicable provisions of Addendum No. _____ to MIL-D-23222A(AS), dated _____." This form shall signed by an authorized representative of the contractor.

b. Technical Data. Documentation on each model of Contractor-acquired SFTI shall contain as a minimum the equipment, nomenclature, model number, operating procedures, applicable preventative maintenance procedures, and physical and performance specifications.

The reporting format and other information details required by 3.19.2.1.2, 3.19.2.1.3 and 3.19.2.1.4 shall be obtained from COMNAVAIRTESTCEN (TS241).

NOTE: When requested, the representatives of the contractor shall confer wish Government personnel at the NAVAIRTESTCEN to reach agreement on. details that are determined to be necessary for inclusion in subsequent revisions to Instrumentation Reports, paragraph 3.19.2.1."

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Paragraph 3.19.3, Military specification aircraft demonstration report - distribution of. Required, except make the following changes to table:

a. In the first block: Delete "3.19.2.1" and substitute "3.19.2.1.1," and under column "Submittal Time and Remarks": Delete "3.19.2.1" and substitute "3.19.2.1.1."



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b. Add paragraphs 3.19.2.1.2, 3.19.2.1.3 and 3.19.2.1.4:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| "3.19 .2.1.2 | Special Flight Test Instrumentation (SFTI) Inventory and Status Report | (b) Attn: SFTIP (TS241) | Info | 1 | Nonrepro | Recurring | Submit by 1 May each year |
| 3.19.2.1.3 | Special Flight Test Instrumentation (SFTI) Excess Equipment Report | (b) Attn: SFTIP (TS241) | Action | 1 | Nonrepro | Recurring | Submit at the time SFTI becomes excess |
| 3.19.2.1.4 | Special Flight Test Instrumentation (SFTI) Requisition and Technical Data | (b) Attn: SFTIP (TS241) | Info | 2 | Nonrepro | Recurring | Submit on the 15th of each month following the month that SFTI was received by the contractor. Reports not required when no SFTI was received during previous month" |

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